

BAYANBULAK, China

WMO: 515420

Lat: 43.0336N

Lon: 84.1491E

Elev: 8068

StdP: 10.89

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-36.7	-32.5	-42.2	0.6	-36.3	-37.8	0.9	-32.3	25.4	2.5	20.6	2.6	1.2	340	0.818

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.9	70.6	51.6	68.0	50.5	65.4	49.6	54.1	66.1	52.6	64.1	51.2	62.2	7.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.5	71.0	57.8	47.7	66.4	56.2	46.3	62.9	55.3	26.1	66.3	24.9	64.5	24.0	62.4	61.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	25.9	22.3	18.8	DB	-40.2	76.4	5.9	2.6	-44.4	78.3	-47.9	79.9	-51.2	81.4	-55.6	83.3
(5)				WB	-40.2	56.8	5.4	2.0	-44.1	58.3	-47.3	59.5	-50.3	60.6	-54.3	62.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.1	-14.1	-5.8	14.9	34.7	43.2	49.9	53.1	51.4	43.8	31.9	14.2	-5.9	(6)
(7)		DBStd	24.93	10.55	11.04	12.19	8.01	5.85	4.50	4.04	4.51	5.62	8.09	11.97	10.64	(7)
(8)		HDD50	8982	1987	1564	1088	460	218	55	14	35	193	560	1075	1733	(8)
(9)		HDD65	14202	2452	1984	1553	910	675	454	368	422	636	1025	1525	2198	(9)
(10)		CDD50	257	0	0	0	0	8	51	111	79	8	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	21	0	0	0	0	0	0	18	3	0	0	0	0	(12)
(13)		CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.4	3.3	4.3	6.6	9.1	8.8	7.5	7.1	6.8	6.9	6.9	5.5	4.1	(14)
(15)	Precipitation	PrecAvg	10.4	0.1	0.1	0.2	0.4	1.2	2.4	2.4	2.0	1.1	0.2	0.1	0.1	(15)
(16)		PrecMax	14.0	0.5	0.4	0.7	0.8	2.6	5.8	6.3	4.3	2.4	0.6	0.7	0.4	(16)
(17)		PrecMin	8.3	0.0	0.0	0.0	0.1	0.4	0.9	0.8	0.5	0.4	0.0	0.0	0.0	(17)
(18)		PrecStd	1.4	0.1	0.1	0.2	0.2	0.6	1.1	1.1	0.9	0.6	0.2	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.9	25.2	47.0	63.9	67.7	70.3	76.4	73.5	69.2	59.0	45.6	22.9	(19)
MCWB			14.7	20.7	34.6	43.3	46.4	50.3	53.7	52.9	48.2	41.6	33.5	19.7	(20)	
2%		DB	11.1	19.2	41.1	58.3	63.7	67.3	71.2	70.2	65.0	53.9	39.5	17.9	(21)	
		MCWB	9.4	16.4	32.5	41.0	45.2	49.4	53.1	51.6	47.0	39.5	30.4	15.3	(22)	
5%		DB	6.4	14.9	36.4	53.4	60.4	64.7	68.0	67.3	61.6	50.1	34.7	14.1	(23)	
		MCWB	4.9	12.8	29.7	38.9	44.1	48.8	51.9	50.8	45.8	37.5	28.3	12.1	(24)	
10%		DB	1.6	10.7	32.4	49.2	56.9	61.6	64.7	64.3	58.2	46.3	29.9	9.7	(25)	
		MCWB	0.5	9.2	27.8	37.1	42.7	47.8	51.1	49.9	44.4	35.7	25.1	8.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.8	21.1	36.6	45.2	49.0	53.3	57.2	55.7	50.2	43.0	35.2	19.4	(27)
MCD B			16.7	24.6	45.0	59.8	62.6	64.6	68.4	67.8	64.9	55.6	42.0	22.3	(28)	
2%		WB	9.5	16.4	33.1	42.2	46.9	51.5	54.9	53.7	48.2	40.7	31.4	15.5	(29)	
		MCD B	11.1	19.0	39.5	55.3	59.9	62.6	67.3	65.7	61.8	51.8	38.4	17.8	(30)	
5%		WB	5.0	12.9	30.5	40.0	45.3	50.2	53.3	52.1	46.6	38.7	28.3	12.1	(31)	
		MCD B	6.4	14.9	35.8	51.1	57.3	61.0	64.8	63.7	59.1	48.2	34.6	14.2	(32)	
10%		WB	0.5	9.1	28.0	38.0	43.7	49.0	51.8	50.7	45.1	36.5	25.1	8.2	(33)	
		MCD B	1.6	10.6	32.5	47.4	54.7	59.3	63.0	61.8	56.6	44.7	29.8	9.7	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	18.9	20.8	21.6	21.5	21.7	20.1	20.9	22.9	24.8	24.7	20.4	18.0	(35)
MCDBR			21.3	21.6	23.0	31.5	29.9	27.3	28.2	30.6	32.8	31.1	27.2	21.0	(36)	
MCWBR			19.5	19.0	15.1	16.1	14.3	12.3	12.8	14.3	16.5	17.7	18.2	18.3	(37)	
5% WB		MCDBR	21.3	21.4	20.7	26.0	25.1	21.9	23.7	24.9	28.6	27.4	26.1	20.8	(38)	
		MCWBR	19.5	18.8	14.1	14.0	12.3	10.4	11.3	12.2	14.6	15.5	17.6	18.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.239	0.275	0.326	0.367	0.362	0.373	0.355	0.348	0.331	0.298	0.264	0.234	(40)	
(41)		taud	2.393	2.299	2.201	2.119	2.214	2.243	2.341	2.355	2.358	2.379	2.403	2.428	(41)	
(42)		Ebn at Noon	292	294	289	284	288	284	288	287	285	282	278	285	(42)	
(43)		Edh at Noon	24	31	40	47	44	43	39	37	34	29	24	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	722	1036	1391	1637	1817	1849	1861	1735	1444	1040	720	596	(44)	
(45)		RadStd	35	48	90	121	115	117	109	98	78	83	48	40	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page